

Testing, Acceptance & Anti-Bluff QA Strategy (constitution §11.4.169)

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Reconciled (§11.4.35, 2026-06-26): §2 now states the test-type set consistently as **16 bundled families / 18 enumerated codes** (resolving the prior "Sixteen types" prose vs the 18 test_type CHECK codes in the v08 coverage-ledger schema). §7.2 (DoD = nine ACs, AC1–AC9) is unchanged and is the canonical authority the v08 child docs were reconciled to.

Master technical specification — document 10 of the HelixVPN set. This is the **canonical QA contract**: it defines, once, the §11.4.169 test-type taxonomy, the captured-evidence model, the harnesses, the per-phase acceptance gates, and the coverage-ledger schema that the Phase-0 and Phase-1 work-breakdown documents ([06](#), [07](#)) reference by abbreviation. Every other document's per-item Tests: column draws from the closed vocabulary fixed *here* (§2). It is **spec-only** — it describes *how "works for the end user" is mechanically proven*, never the product itself (2–3 refinement passes follow).

Authoritative companions cited inline: data plane [01]; control plane [02]; client core + UI [03]; security/privacy/PKI [04]; repo + ecosystem [05]; Phase-0 WBS [06]; Phase-1 WBS [07]. Primary research: **[04_P0]** (04_VPN_CLD/HelixVPN-Phase0-Spike.md), **[04_P1]** (Phase1-MVP), **[04_ARCH]**, **[04_UI]** (HelixVPN-helix-ui-Flutter.md), **[05_YBO]**, **[SYNTHESIS]**, LLM analyses **[02_QWN]** **[01_DSK]** **[11_MST]** **[10_KMI]** **[07_GMI]**, external facts **[research-masque]** (RFC 9297/9298/9221), **[research-boringtun]**, **[research-quinn]**, **[research-wireguard]**.

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0. Doctrine: the anti-bluff covenant applied to a VPN

A VPN is the worst possible product to bluff-test, because **every failure mode is silent and invisible to a green summary line**. A kill-switch that "passes" but leaks one DNS query deanonymises the user. A tunnel that "connects" but routes plaintext is worse than no VPN. A policy edit that "applies" but leaves a stale AllowedIPs grants access that was revoked. The §11.4 covenant's founding mandate — *"execution of tests MUST guarantee ... full usability by end users"* — is not a slogan here; it is the only thing standing between the product and a deanonymisation incident.

Therefore the operative bar for **every** HelixVPN test is not "the assertion returned true" but **"a packet, a pcap, a counter delta, or a rendered frame — captured during execution — proves the user-visible security property held."** Five HelixVPN-specific bluff classes are forbidden by name (each maps to a constitution clause and is mechanically guarded in §5):

Bluff class	What it looks like	Forbidden by	Guarded in
B1 — config-only PASS	asserting AllowedIPs contains the route, never sending a packet	§11.4.69	§5.3 E2E, §5.7 SEC
B2 — absence-of-error PASS	"kill-switch enabled, no error" with no leak-capture	§11.4.68 / .107	§5.7 SEC, AC7 (§7.2)
B3 — wrong-plane PASS	dumpsys/policy field says "connected" while no PCM/packet flows	§11.4.107(2), §11.4.68 §11.4.108/.139	§5.3, §5.13 §8, §9

Bluff class	What it looks like	Forbidden by	Guarded in
B4 — stale-state PASS	validating against a previous deploy's running daemon		
B5 — unvalidated-analyzer PASS	the pcap/OCR analyzer passes its own golden-bad fixture	§11.4.107(10)	§3.3, §5.14

Every PASS in this strategy flows through `ab_pass_with_evidence <desc> <evidence_path>` (§11.4.69); bare `ab_pass` is a release blocker. A FAIL that crashes for a script-internal reason (set `-u undefined var`, a broken `tshark` filter) is **equally forbidden** (§11.4.1) — the harness sources a shared anti-bluff lib (§9) so script bugs surface at the lib layer, not as fake product FAILs.

1. §11.4.169 — the universal test-type mandate

§11.4.169 (the §11.4.27 enumeration, promoted to a per-item closure obligation) requires that **every workable item declares the closed set of test types it must pass before it may close**, and that the *absence* of any warranted type is never silent — it is an explicit `NOT_APPLICABLE: <reason>` or an honest §11.4.3 SKIP: `<closed-set-reason>`, never an omission. This document fixes that closed set, its HelixVPN instantiation, and its evidence shape. The `Tests:` columns in [06] and [07] are projections of §2 below.

Composition (per §11.4.169): §11.4.4(b) four-layer (pre-build gate + post-build

- runtime-on-clean-target + paired §1.1 mutation) applies to *every* item closure; §11.4.50 determinism (N=3 normal / N=10 cycle-validation, identical evidence hashes) applies to every PASS; §11.4.132 risk-ordering governs run order; §11.4.149 records each run in the item's `test_diary`; §11.4.135 registers a permanent regression guard for every closed defect.

2. The closed test-type taxonomy (canonical)

This is the **single source** for the abbreviations used project-wide — **16 bundled families / 18 enumerated codes** (the families below expand to the 18 `test_type` CHECK codes of the §6 coverage-ledger schema: UI/UX and CHAL/HQA each render as one family but distinct codes). DDOS and SCALE are MVP-NOT_APPLICABLE by design (single-node self-host) and re-arm in Phase 2 (HA / managed).

Abbrev	§11.4.169 type	HelixVPN floor expectation	Primary harness	MVP status
UNIT	unit	pure logic: policy compiler, IPAM allocator, MapResponse delta diff, MASQUE framing, status FSM.	cargo test, go test, dart test	required

Abbrev	§11.4.169 type	HelixVPN floor expectation	Primary harness	MVP status
		Mocks allowed only here (§11.4.27).		
INT	integration	real Postgres+Redis+edge booted on-demand via containers submodule (§11.4.76); no mocks .	testcontainers via containers/pkg/boot	required
E2E	end-to-end	real control plane drives the netns rig; curl reaches an authorized LAN host through the real tunnel.	netns + nftables + netem rig [04_P0 §3]	required
FA	full-automation	self-driving, re-runnable –count=3, zero human-in-loop (§11.4.98).	make spike / make qa one-shot	required
SEC	security	RLS cross-tenant denial, key-never-leaves-FFI, mTLS, kill-switch, DNS-leak, DPI-evasion, secret-leak audit.	netns + pcap + security submodule	required
DDOS	DDoS / load-flood	handshake-flood + volumetric flood; edge fail-static + rate-limit holds.	iperf3 flood + custom WG-init flood	NOT_APPLICABLE: single-node-selfhost (Phase 2)
STRESS	stress	sustained load (≥ 100 iters / ≥ 30 s), ≥ 10 parallel, boundary inputs (§11.4.85).	stress_chaos.sh ab_stress_*	required
CHAOS	chaos	mid-flight SIGKILL / Redis drop / iface flap / partial write; recovery restores consistent state (§11.4.85).	stress_chaos.sh ab_chaos_*	required
CONC	concurrency / atomicity	concurrent IPAM allocs / enrollments / policy edits; no lost update, no double-allocation.	Go –race + Postgres txn harness	required
RACE	race / deadlock	data-race + deadlock detection on the hot paths (reconcile loop, edge verdict swap).	cargo +nightly loom + Go –race + tsan	required
MEM	memory	iOS NEPacketTunnelProvider RSS under the ~15 MB historical ceiling, $\geq 30\%$ headroom; no leak over 24 h soak.	Instruments / / proc / dumsys meminfo	required (G3-critical)
BENCH	benchmarking			required

Abbrev	§11.4.169 type	HelixVPN floor expectation	Primary harness	MVP status
		throughput, CPU-per-Gbps, handshakes/sec, p99 framing latency.	<code>bench.sh</code> [04_P0 §8] + <code>criterion</code>	
PERF	performance	latency p50/p99 vs the SLO budget (§7.2); Go GC-tail watch on hot paths.	<code>bench.sh</code> + histogram metrics	required
SCALE	scaling	N simulated agents holding streams; convergence + bounded memory / 24 h.	agent-fuzzer + coordinator soak	partial (SLO4 soak); full Phase 2
UI / UX	UI / UX	Flutter widget + golden tests; flow walkthrough (enroll→connect→reach) with vision verdict.	<code>flutter test</code> + panoptic drive	required
REC	recorded-evidence	window-scoped MP4 (§11.4.154/.155) + media-validation pipeline verdict (§11.4.163).	panoptic capture → <code>vision_engine</code>	required (per user-visible item)
CHAL / HQA	Challenge / HelixQA	a challenges / <code>helix_qa</code> bank entry scoring PASS only on captured evidence (§11.4.27/.107).	challenges + <code>helix_qa</code> submodules	required (DoD ACs)

Why DDOS/SCALE defer. [04_P1 §11] scopes the MVP to a single rootless host with no public multi-tenant surface; a DDoS suite at MVP would test a threat the deployment topology does not yet present (§11.4.3 topology-SKIP). The seam is built now (rate-limit token buckets [04 §T9], stateless fail-static edge [01 I3]) and the suite is authored as a **parked Phase-2 bank** so it re-arms mechanically when HA lands, never as an omission.

3. The captured-evidence model (§11.4.5 / .69 / .107)

3.1 Evidence taxonomy mapped to §11.4.69 sink-side classes

Every HelixVPN feature maps to one §11.4.69 sink-side evidence class; a PASS cites an artifact whose *shape* the class fixes. This table is the contract `vision_engine` and the challenges banks score against.

Feature class	§11.4.69 class	Captured-evidence shape (the artifact <code>ab_pass_with_evidence</code> cites)
tunnel carries traffic	network_throughput network_connectivity	<code>iperf3 -J</code> JSON with <code>sum_received.bits_per_second > 0</code> ; pcap with ≥ 1 decrypted inner packet

Feature class	§11.4.69 class	Captured-evidence shape (the artifact <code>ab_pass_with_evidents</code> cites)
authorized reachability		<code>curl HTTP 200 body hash</code> from the overlay netns + pcap SYN→SYN-ACK
default-deny	<code>network_connectivity</code> (negative)	pcap showing SYN out, zero SYN-ACK; edge nft/eBPF verdict-map counter drop++
kill-switch / DNS-leak	<code>wifi_link</code> / negative	host-side pcap during forced drop: zero non-loopback packets, zero :53 egress
transport escalation	<code>network_connectivity</code>	<code>StatusReport.transport=="masque-h3"</code> + tshark http3 classification, no WG signature
policy reconcile latency	counter delta	<code>helix_reconcile_seconds</code> p99 histogram bucket < 1 s
revoke latency	counter delta	<code>revoke-event-ts</code> → <code>edge-peer-removed-ts</code> timing CSV, p99 < 1 s
key-never-leaves	negative	FFI boundary scan + log scan: zero private-key bytes cross FFI / hit any log
iOS NE memory	MEM	<code>Instruments.trace / footprint</code> RSS series, peak < ceiling, ≥30% headroom
UI connect flow	REC	window-scoped MP4 (§11.4.154) + <code>vision_engine</code> OCR verdict on the <code>ConnectButton</code> state

3.2 §11.4.107 liveness battery (the no-frozen-frame rules, applied)

The §11.4.107 mandate — *a single captured frame/sample is not proof* — instantiates for HelixVPN as:

1. **Liveness over a window**, not a point. Throughput is sampled over ≥10 s; a one-shot ping reply is a pre-filter, never the proof (§11.4.107(1)).
2. **Independent counter-advance**. Goodput (`iperf3`) **and** an independent kernel WG counter (`wg show <if> transfer rx/tx bytes` advancing) must *both* move — a flat WG counter with moving `iperf3` means a decoy/loopback path (§11.4.107(2)).
3. **Not-stale cross-check**. After a transport escalation, the post-escalation pcap's first inner packet must not be a replay of the pre-escalation flow's last packet (§11.4.107(4)).
4. **Loading is a distinct state**. A handshake in progress is `Connecting`, not `Connected`; the FSM must reach `Connected` before liveness is judged (§11.4.107(3)); `timeout+reachable` ⇒ FAIL, `timeout+unreachable` ⇒ SKIP.
5. **Metamorphic relations** where there is no golden source: same LAN host reached over WG-UDP vs MASQUE must return the **same body hash**; a paused tunnel stops the counter; 2× clients ⇒ ~2× aggregate goodput (§11.4.107(8)).

3.3 Self-validated analyzers (§11.4.107(10)) — defeating B5

Every analyzer (pcap classifier, leak detector, OCR verdict, memory parser) ships with a **golden-good + golden-bad fixture pair** and is wired into the meta-test sweep. An analyzer that passes its golden-bad fixture is itself the bluff and FAILs CM-ANALYZER-SELF-VALIDATED.

```
// tests/analyzers/leak_detector_selfvalidation.rs
    (§11.4.107(10))
#[test] fn golden_good_clean_drop_passes() {
    let v = LeakDetector::new().scan_pcap("fixtures/
        killswitch_clean.pcap");
    assert_eq!(v.verdict, Verdict::Pass);           // a truly-
        clean drop PASSes
}
#[test] fn golden_bad_leaked_dns_fails() {
    let v = LeakDetector::new().scan_pcap("fixtures/
        killswitch_leaked_dns.pcap");
    assert_eq!(v.verdict,
        Verdict::Fail);           // a seeded :53 leak MUST FAIL
    assert!(v.findings.iter().any(|f| f.proto == "dns" && f.dport
        == 53));
}
```

The paired §1.1 mutation for this gate strips the `dport == 53` clause from the detector and asserts `golden_bad_leaked_dns_fails` now *passes the leak* → meta-test FAILs → the mutation is caught. This is the mechanical defeat of B5.

4. The test pyramid + ecosystem topology (§11.4.156 — local, no remote CI)

```
graph TD
    subgraph LOCAL["LOCAL gates only – §11.4.156 remote CI DISABLED"]
        direction TB
        L1["pre-commit hook<br/>UNIT + sh -n + schema-lint + no-log lint + mutation-residue scan"]
        L2["pre-push hook<br/>INT (containers boots PG+Redis) + SEC subset + propagation gates"]
        L3["make test / make qa<br/>full pyramid on demand, backgrounded §11.4.89"]
    end

    subgraph PYRAMID["The HelixVPN test pyramid"]
        direction BT
        U["UNIT – Rust/Go/Dart, mocks here only<br/>(policy compiler, IPAM, delta diff, MASQUE framing)"]
        I["INT – real PG+Redis+edge via containers submodule<br/>RLS · enrollment · reconcile"]
        E["E2E – netns rig: client→gateway→connector LAN<br/>curl reaches authorized host"]
    end
```

```

    SC["SEC · STRESS · CHAOS · CONC · RACE · MEM · BENCH<br/>
>cross-cutting, risk-ordered first §11.4.132"]
    CH["CHAL + HQA – challenges + helix_qa banks<br/>score the 8
MVP–DoD ACs on captured evidence"]
    RE["REC – panoptic capture → vision_engine OCR verdict<br/>
>window-scoped MP4 per user-visible flow"]
    U --> I --> E --> SC --> CH --> RE
end

subgraph INFRA["Test infra via submodules (§11.4.76 – sole
container path)"]
    C1["containers/pkg/boot · health · compose<br/>boots
PG+Redis+gateway on-demand, tears down"]
    C2["challenges – anti-bluff Challenge engine"]
    C3["helix_qa – autonomous QA orchestrator"]
    C4["vision_engine – video/OCR evidence analysis"]
    C5["panoptic – UI capture (web+desktop+mobile drive)"]
    C6["security – DPI/leak/secret tooling"]
end

L1 --> U
L2 --> I
L3 --> E
L3 --> SC
L3 --> CH
L3 --> RE
I -.boots.-> C1
CH -.uses.-> C2
CH -.uses.-> C3
RE -.capture.-> C5
RE -.analyze.-> C4
SC -.uses.-> C6

```

§11.4.156 binding. There is **no active .github/workflows/*.yml or .gitlab-ci.yml** in any HelixVPN repo. The CI workflow file is preserved at **.github/workflows/qa.yml.disabled-local-only** (a name no provider triggers, §11.4.75 Layer-5). Enforcement is the **local** git-hook ritual (§9) + the §11.4.40 pre-tag sweep run by the operator. Every gate in the pyramid is a Makefile target that runs on the developer's host; containers boots infra on demand so no shared runner is required.

5. Per-type strategy, harness, evidence, gate, skeletons

5.1 UNIT

Covers (HelixVPN). Pure deterministic logic where a bug is a logic bug, not an integration bug: the policy compiler (ACL → AllowedIPs + verdict-map, default-deny correctness), the IPAM allocator (overlay /48 ULA carve, collision-free allocation, D4 [SYNTHESIS]), the coordinator delta diff (snapshot→minimal-delta

correctness), MASQUE/HTTP-Datagram framing encode/decode (RFC 9221 [research-masque]), the client status FSM transitions, kill-switch firewall-rule synthesis.

Harness. `cargo test` (Rust core/edge), `go test` (control plane), `dart test` (UI logic). Property tests via `proptest` (Rust) and `gopter` (Go) for the compiler and allocator — these are exactly the components where exhaustive hand-written cases miss the boundary that bites. **Mocks are permitted only at this layer** (§11.4.27).

Evidence. Test output + coverage report; for the compiler a *golden-table* fixture (`acl.yaml` → expected AllowedIPs set) committed as a regression guard (§11.4.135).

Gate. Pre-commit hook (§9). Coverage floor ratchets 70→85→95% over phases (§11.4.50 feature-coverage matrix).

```
// helix-core/crates/helix-policy/src/compile.rs (interface under
    test)
pub trait PolicyCompiler {
    /// Compile a tenant ACL into per-peer AllowedIPs + an edge
    /// verdict map.
    /// MUST be default-deny: an empty/parse-failed ACL yields
    /// ZERO allowed routes.
    fn compile(&self, acl: &Acl, peers: &[PeerId]) ->
        Result<CompiledPolicy, CompileError>;
}

// helix-core/crates/helix-policy/tests/default_deny.rs
#[test] fn empty_acl_grants_nothing() { //
    the security floor
    let c = TailscaleAclCompiler::default();
    let cp = c.compile(&Acl::empty(), &[peer("alice"),
        peer("bob")]).unwrap();
    for p in cp.peers() { assert!(cp.allowed_ips(p).is_empty(),
        "fail-closed violated"); }
}
#[test] fn parse_error_is_fail_closed_not_fail_open() { //
    B-class: never fail-open
    let c = TailscaleAclCompiler::default();
    assert!(matches!(c.compile(&Acl::malformed(), &[]),
        Err(CompileError::Parse(_))));
    // and the caller MUST treat Err as "no routes", proven in the
    INT layer (§5.2)
}
```

5.2 INT — integration (containers submodule infra)

Covers. Every seam that touches real infrastructure: Postgres RLS multi-tenant isolation (a tenant-B query under tenant-A's RLS context returns zero rows), device enrollment end-to-end (OIDC/anon token → device-generated WG keypair, private key never persisted server-side → short-lived mTLS cert issued), event backbone (Redis Streams publish → coordinator consume → delta emitted), the coordinator WatchNetworkMap server-stream (snapshot + deltas, peers already policy-filtered), no-log schema invariant against the live DB.

Harness — the **containers** submodule is the **ONLY** path (§11.4.76). Integration tests boot Postgres + Redis (+ edge where needed) **on-demand** via containers/pkg/boot, run against the *real* services, and tear down — no docker/podman CLI outside pkg/boot/pkg/compose/pkg/health, no mocks (§11.4.27). This is the on-demand-infra invariant: make test boots infra itself, the developer never runs podman by hand.

```
// helix-go/internal/store/rls_integration_test.go
//go:build integration
package store

import (
    "context"; "testing"
    "digital.vasic.containers/pkg/boot"    // §11.4.76 sole
        container seam
    "digital.vasic.containers/pkg/health"
)

func TestRLSCrossTenantDenial(t *testing.T) {
    ctx := context.Background()
    infra, err := boot.Up(ctx, boot.Spec{Postgres: true, Redis:
        true, Rootless: true}) // §11.4.161
    if err != nil { t.Fatalf("boot infra: %v", err) }
    t.Cleanup(func() { _ = infra.Down(ctx) }) //
        §11.4.14 quiescence
    if err := health.WaitReady(ctx, infra, health.Postgres); err !=
        nil { t.Fatal(err) }

    db := mustConnect(t, infra.PostgresDSN())
    seedTenant(t, db, "tenant-A", "device-a1")
    seedTenant(t, db, "tenant-B", "device-b1")

    // Under tenant-A's RLS role, tenant-B's devices MUST be
        invisible (real DB, no mock).
    rows := queryAs(t, db, "tenant-A", `SELECT id FROM devices`)
    requireContains(t, rows, "device-a1")
    requireNotContains(t, rows, "device-b1") //
        captured: rowset = evidence (§11.4.5)
    // paired §1.1 mutation: disable the RLS policy → this test
        MUST FAIL (gate CM-RLS-ENFORCED)
}
```

Evidence. The captured rowset / stream transcript; for the no-log invariant the schemalint report. **Gate.** Pre-push hook + make test.

```
-- the no-log invariant the INT + schema-lint layers enforce (§04,
    [02 §2.4])
-- CI schema-lint FAILS the build if ANY durable connection/
    traffic/packet table appears.
-- Allowed: aggregate counters only.
CREATE TABLE traffic_counters (                -- ALLOWED:
    aggregate, no per-flow rows
    tenant_id uuid NOT NULL,
    window_start timestampz NOT NULL,
```

```

    rx_bytes    bigint NOT NULL,                -- SUM over window,
              not per-connection
    tx_bytes    bigint NOT NULL,
    PRIMARY KEY (tenant_id, window_start)
);
-- FORBIDDEN (schemalint paired §1.1 mutation seeds this → gate
  MUST FAIL):
-- CREATE TABLE connections (src_ip inet, dst_ip inet, started_at
  timestampz, ...);

```

5.3 E2E — end-to-end (netns rig)

Covers. The whole reachability slice on one box: a *real* control plane issues a NetworkMap, a *real* helix-core client builds the tunnel, traffic crosses client → gateway → connector LAN, and curl http://10.10.0.20/ returns the hello page. Plus the *negative* E2E (default-deny: a SYN out, zero SYN-ACK) and the DPI-block E2E (plain WG blocked → MASQUE escalation → reachability preserved). The rig is the reproducible substrate for AC2/AC3/AC4 (§7.2) and survives Phase-0→1.

Harness — Linux netns + nftables-DPI + tc-netem [04_P0 §3, 06 §HVPN-P0-022]. Three processes (client, gateway, connector) in namespaces on one host; an nft table simulates a DPI UDP block; tc netem injects loss/delay for resilience.

```

# rig/netns_up.sh – connector-side "private LAN" simulated with a
  netns [06 HVPN-P0-023]
set -euo pipefail
ip netns add lanA
ip link add veth-host type veth peer name veth-lanA
ip link set veth-lanA netns lanA
ip -n lanA addr add 10.10.0.20/24 dev veth-lanA && ip
  -n lanA link set veth-lanA up
ip netns exec lanA python3 -m http.server 80 --bind 10.10.0.20
  & # the hello service (sink)

# rig/dpi_block.sh – DPI/censorship sim for the AC4 escalation E2E
nft add table inet dpi
nft add chain inet dpi fwd '{ type filter hook forward priority
  0; }'
nft add rule inet dpi fwd udp dport 51820 drop          # kill plain
  WireGuard
nft add rule inet dpi fwd udp dport 443 accept          # allow
  MASQUE/H3

# rig/impair.sh – loss/jitter for the resilience E2E (§11.4.85
  adjacency)
tc qdisc add dev "$IFACE" root netem loss 5% delay 40ms 10ms

# rig/test_reach.sh – the E2E assertion (positive + negative),
  §11.4.50 N=3 deterministic
run_reach() {
  expected: pass|deny
  pcap="qa-results/e2e/$(date +%s)_reach.pcap"
  ip netns exec client tcpdump -i overlay0 -w "$pcap" & TPID=$!
}

```

```

body=$(ip netns exec client curl -s --max-time 5 http://
10.10.0.20/ || true)
kill "$TPID" 2>/dev/null
if [ "$1" = pass ]; then
    [ -n "$body" ] || ab_fail "B1: no body – config-only PASS
    forbidden"
    grep -q "SYN-ACK" <(tshark -r "$pcap" -Y
    'tcp.flags.syn==1 && tcp.flags.ack==1') \
    && ab_pass_with_evidence "authorized reach" "$pcap"
else
    [ -z "$body" ] || ab_fail "default-deny breached: got a body"
    tshark -r "$pcap" -Y 'tcp.flags.syn==1 && tcp.flags.ack==1' |
    grep -q . \
    && ab_fail "deny breached: SYN-ACK present" \
    || ab_pass_with_evidence "default-deny (no SYN-ACK)" "$pcap"
fi
}

```

Evidence. pcap + curl body hash + iperf3 -J CSV ($\geq 80\%$ bare-link for G1).

Gate. make test E2E stage; AC2/AC3/AC4 release gates (§7.2). Cleanup is non-negotiable — trap 'rig/netns_down.sh' EXIT leaves the host quiescent (§11.4.14).

5.4 FA — full-automation

Covers. The §11.4.98 invariant: **every** E2E / integration / Challenge must be re-runnable end-to-end with **zero human intervention** after startup. For HelixVPN this means the whole MVP-DoD demo (helixvpnctl init → enroll connector+client → reach → deny → escalate → revoke) runs as one command, N=3 consecutively, with self-cleaning state and identical PASS each time (§11.4.50). The single forbidden human touch-point is credential bootstrap *outside* execution (a .env token, §11.4.10).

Harness. make spike (Phase 0) and make qa (Phase 1) are the one-shot drivers; the latter dispatches helix_qa's autonomous session (§5.6). Determinism is mechanical: ab_run_n_times <name> 3 <fn> loops, captures an evidence-hash per iteration, and FAILs if any hash or exit code diverges — there is no "first-pass-was-a-flake" path (§11.4.50).

```

# the §11.4.50 determinism wrapper every FA suite uses
ab_run_n_times() {
    local name="$1" n="$2" fn="$3" h prev=""
    for i in $(seq 1 "$n"); do
        "$fn" "$i"; h=$(sha256sum "qa-results/$name/run_$i/
        evidence.json" | cut -d' ' -f1)
        [ -z "$prev" ] || [ "$h" = "$prev" ] || ab_fail "non-
        deterministic: run $i hash diverged"
        prev="$h"
    done
    ab_pass_with_evidence "$name x$n deterministic" "qa-results/
    $name/"
}

```

5.5 CHAL — Challenges (challenges submodule banks)

Covers. The eight MVP Definition-of-Done acceptance criteria (AC1–AC9, §7.2) are authored as **executable Challenges** in the challenges submodule — the constitution's anti-bluff Challenge layer (§11.4.27(B)/.5/.69). A Challenge differs from a test: it scores PASS **only** on captured evidence matching the §3.1 shape for that feature class, and is graded by the challenges engine, not by the test's own exit code. This is the structural defence against B1–B3: the Challenge engine re-reads the pcap/counter/recording, it does not trust the harness's claim.

Harness. challenges submodule (digital.vasic.challenges, Go) consumed via replace in dev / pinned SHA in release [05 §6.1]. A bank file maps each AC to its on-rig driver + the evidence-class assertion.

```
# challenges/banks/helixvpn_mvp_dod.yaml (illustrative – one
    entry per AC)
bank: helixvpn-mvp-dod
challenges:
  - id: HVPN-CHAL-AC2-authorized-reach
    feature_class: network_connectivity           # §11.4.69
    driver: rig/test_reach.sh pass
    evidence:
      kind: pcap
      assert: "frame.tcp.flags.syn==1 && frame.tcp.flags.ack==1
        from 10.10.0.20"
      and_body_hash_nonempty: true
    score: pass_only_if_evidence_matches         # never trust exit
      code alone
  - id: HVPN-CHAL-AC7-killswitch-no-leak
    feature_class: killswitch_negative
    driver: rig/killswitch_drop.sh
    evidence:
      kind: pcap
      assert_zero: "ip && not loopback"          # zero plaintext
        egress
      assert_zero_dns: "udp.port==53"           # zero DNS leak
      self_validated: true                      # golden-bad
      seeded-leak pcap MUST score FAIL (§3.3)
```

Evidence + Gate. The Challenge engine's per-AC verdict JSON under docs/qa/<run-id>/ (§11.4.83); make qa is the gate; a release tag is blocked while any DoD Challenge is non-PASS.

5.6 HQA — HelixQA autonomous sessions

Covers. helix_qa is the **autonomous QA orchestrator** (§11.4.27/.107/.158): it drives the full MVP-DoD acceptance set end-to-end with no human, detects and respawns crashed sub-runs (§11.4.147), generates tickets for findings into the workable-items DB, and emits the real-time sync channel (§11.4.116 JSONL event stream + atomic status snapshot) that the conductor tails. It is the §11.4.165 independent-verification agent for the runtime layer — structurally separate from the code author.

Harness. go run ./submodules/helix_qa/cmd/orchestrator --suite mvp-dod ([05 Makefile qa:]). It consumes the challenges banks (§5.5), boots infra via containers (§5.2), drives panoptic for UI flows (§5.14), and feeds frames to vision_engine (§5.14). Each verdict event carries its evidence path (§11.4.116 — a PASS with no evidence path is a contradiction → treated as FAIL).

```
// the §11.4.116 sync channel helix_qa emits (qa-results/helix_qa/
events.jsonl, append-only)
{"ts":"2026-...", "ev":"session_start", "suite":"mvp-
dod", "build":"<artifact-md5>"}
{"ts":"...", "ev":"challenge_start", "id":"HVPN-CHAL-AC2-authorized-
reach"}
{"ts":"...", "ev":"evidence", "id":"HVPN-CHAL-AC2-...", "path":"qa-
results/.../reach.pcap"}
{"ts":"...", "ev":"verdict", "id":"HVPN-CHAL-
AC2-...", "result":"PASS", "evidence":"qa-results/.../reach.pcap"}
```

Evidence + Gate. The events stream + per-AC verdict; make qa exit status gates the release. HelixQA's own meta-test plants a known-broken AC and asserts the session reports FAIL (§11.4.32 self-meta-test).

5.7 SEC — security

Covers. The security invariants from [04 §13], each a falsifiable captured-evidence row:

Invariant	Captured-evidence proof
S1 default-deny	negative E2E (§5.3): SYN out, zero SYN-ACK, edge drop++
S2 key-never-leaves	FFI boundary scan + log scan: zero private-key bytes cross FFI or hit any log [04 §13]
S3 mTLS device cert	handshake pcap shows client cert; expired/revoked cert ⇒ stream refused
S4 kill-switch	host pcap during forced tunnel drop: zero non-loopback egress [04, §11.4.68]
S5 DNS-leak	host pcap during drop + reconnect: zero :53 to a non-tunnel resolver
S6 DPI-evasion	tshark classifies MASQUE flow as HTTP/3, no WG signature, SNI = decoy host
S7 wire-fingerprint	passive observer cannot distinguish HelixVPN MASQUE from generic H3 (entropy + timing)
S8 secret-leak audit	git ls-files xargs grep -l <secret> + git log -S<secret> empty (§11.4.10/.10.A)
S9 RLS cross-tenant	§5.2 INT rowset isolation

Harness. netns + pcap for S1/S3–S7; the security submodule for the DPI/leak/secret tooling [05 §6 row]; FFI scan tool for S2. Each analyzer is self-validated (§3.3). **Gate.** Pre-push hook (secret audit + RLS subset) + make qa SEC stage. The secret-leak audit is also a §11.4.10.A pre-store audit run before any credential lands.

```
# rig/killswitch_drop.sh - S4+S5, the anti-bluff kill-switch test
# (defeats B2)
set -euo pipefail
pcap="qa-results/sec/$(date +%s)_killswitch.pcap"
ip netns exec client tcpdump -i any -w "$pcap" & TPID=$!
ip netns exec client curl -s --max-time 30 http://10.10.0.20/ >/
dev/null & # traffic in flight
sleep 2
rig/force_tunnel_drop.sh # core FSM ->
Blocked, firewall seals
sleep 5
ip netns exec client nslookup example.com 2>/dev/null || true #
try to leak a DNS query
kill "$TPID" 2>/dev/null
# PASS only if, after the drop, ZERO non-loopback packets AND ZERO
:53 left the host:
LEAK=$(tshark -r "$pcap" -Y 'frame.time_relative>2 && ip && not
(ip.addr==127.0.0.1)' | wc -l)
DNS=$(tshark -r "$pcap" -Y 'frame.time_relative>2 &&
udp.port==53' | wc -l)
[ "$LEAK" -eq 0 ] && [ "$DNS" -eq 0 ] \
&& ab_pass_with_evidence "kill-switch sealed, no DNS leak"
"$pcap" \
|| ab_fail "LEAK=$LEAK DNS=$DNS - plaintext/DNS escaped the
seal"
```

5.8 DDOS — denial-of-service / load-flood

Covers (Phase 2; seam built MVP). Two threats: (a) **handshake flood** — a torrent of WG/MASQUE init packets must not exhaust the gateway (rate-limited via Redis token buckets [04 §T9]); (b) **volumetric flood** — the stateless edge must fail-static (drop, never crash; rootless auto-restart [01 I3]). At MVP this is NOT_APPLICABLE: single-node-selfhost (§11.4.6) and **authored as a parked bank** that re-arms in Phase 2 when the public multi-tenant surface exists.

Harness (parked). iperf3 N-client flood + a custom Noise-IK-init flood generator; assert p99 handshake latency for a *legitimate* client stays bounded while the flood runs, and the edge process never OOM/crashes (counter wg_init_dropped rises, helix_edge_up stays 1). **Evidence.** legit-client latency CSV + edge liveness counter during flood. **Gate.** Phase-2 release gate; MVP records the SKIP reason in the coverage ledger (§6).

5.9 STRESS + CHAOS

Covers. Resilience of every fix (§11.4.85 mandate — happy-path-only is a bluff at the resilience layer). **STRESS:** sustained reconcile churn (≥ 100 policy edits / ≥ 30 s), ≥ 10 concurrent enrollments, boundary inputs (empty ACL, max-peer map, off-by-one overlay range). **CHAOS:** mid-transfer SIGKILL of the edge (tunnel recovers, no plaintext leak during the gap), Redis drop mid-reconcile (coordinator degrades gracefully, no lost delta), interface flap (client roams, tunnel re-establishes < 3 s), partial-write of the network-map snapshot (recovery restores a consistent map, never a torn one), connector process kill (peer drops from every map within the convergence budget).

Harness. stress_chaos.sh exposing ab_stress_run, ab_stress_concurrent, ab_chaos_kill_pid_during, ab_chaos_drop_network_during, ab_chaos_iface_flap_during — each composing with ab_pass_with_evidence and a trap '...' EXIT cleanup (§11.4.14 — corrupt-restore / process-restart mandatory).

```
# CHAOS: kill the edge mid-transfer, assert recovery AND no leak
        during the gap (§11.4.85)
ab_chaos_kill_pid_during "$(edge_pid)" 'ip netns exec client
    iperf3 -t 20 -c 10.10.0.20' \
    --on-kill 'sleep 3' \
    --assert 'tunnel_reestablished_within 3s &&
        killswitch_sealed_during_gap' \
    --evidence "qa-results/chaos/edge_kill_$(date +%s)"/"
# the assertion re-reads the gap pcap: zero plaintext egress while
    the edge was dead (composes S4)
```

Evidence. latency.json (p50/p95/p99), recovery_trace.log, state_delta_snapshot.json, gap-pcap. **Gate.** make test STRESS/CHAOS stage; every closed defect registers a permanent CHAOS guard (§11.4.135).

5.10 CONC — concurrency / atomicity

Covers. The places where two callers race over shared state: concurrent IPAM allocation (no two devices get the same overlay IP — a Postgres SELECT ... FOR UPDATE / unique-constraint contract), concurrent enrollment of the same device, concurrent policy edits to the same tenant (last-writer-wins with a version guard, no lost update), concurrent WatchNetworkMap streams receiving deltas while a new peer enrolls (each stream converges to the same final map). The mandate: any method reachable from ≥ 2 goroutines/threads is safe for rapid consecutive calls (project principle 3).

Harness. Go test spawning N goroutines hammering the allocator against real Postgres (via containers), asserting **exactly** N distinct IPs and zero constraint violations; a property test that the final map is invariant to delta arrival order.

```
func TestIPAMNoDoubleAllocationUnderConcurrency(t *testing.T)
{ // CONC + atomicity
    infra := mustBootPG(t) // containers
    submodule (§5.2)
    alloc := NewIPAMAllocator(infra.PostgresDSN(),
        mustParseULA("fd7a:115c:a1e0::/48"))
    const N = 64
    got := make([]netip.Addr, N); var wg sync.WaitGroup
    for i := 0; i < N; i++ { wg.Add(1); go func(i int){ defer
        wg.Done();
        a, err := alloc.Allocate(ctx, deviceID(i));
        requireNoErr(t, err); got[i] = a }(i) }
    wg.Wait()
    requireAllDistinct(t, got) // captured: the
        N-set is the evidence
    // paired §1.1 mutation: drop the FOR UPDATE lock → a
        collision appears → gate FAILs
}
```


Evidence. the allocated-set + Postgres constraint-violation count (must be 0). **Gate.** make test CONC stage; run with Go -race (overlaps §5.11).

5.11 RACE — race-condition / deadlock

Covers. Data races and deadlocks on the hot paths: the client reconcile loop (network-map apply under a concurrent status read — no torn read of AllowedIPs), the edge verdict-map hot-swap (a policy update swaps the verdict map while packets are being classified — no use-after-free, no half-applied map), the Rust core's tick() keepalive/rekey timer under concurrent send/recvd, the Go coordinator's in-memory topology graph under concurrent stream fan-out.

Harness. Rust: cargo +nightly test --features loom (exhaustive interleaving of the reconcile/verdict-swap critical sections) + ThreadSanitizer on the FFI boundary. Go: go test -race on the coordinator + edge. The §11.4 principle 2 (no blocking op inside a held lock) is asserted by a loom test that the verdict-swap never holds the classify lock across an allocation.

```
// helix-edge/tests/verdict_swap_loom.rs - no torn read, no
//                          deadlock under interleaving
#[test] fn verdict_swap_is_atomic_under_loom() {
    loom::model(|| {
        let vm =
            Arc::new(ArcSwap::from_pointee(VerdictMap::empty()));
        let (vm1, vm2) = (vm.clone(), vm.clone());
        let t = loom::thread::spawn(move || {
            vm1.store(Arc::new(VerdictMap::allow("10.10.0.0/24"))); });
        let v =
            vm2.load(); // classifier reads
                        concurrently
        assert!(v.is_consistent()); // never a half-
        applied map (no torn read)
        t.join().unwrap();
    });
}
```

Evidence. loom/tsan/-race clean report (zero findings). **Gate.** make test RACE stage; a -race finding is a release blocker (B-class latent bug per §11.4.145 angle 3).

5.12 MEM — memory (iOS NE soak)

Covers — the make-or-break platform constraint (Phase-0 gate G3 [04_P0]). iOS NEPacketTunnelProvider runs under a hard memory ceiling (~15 MB historical [SYNTHESIS §5]); the entire decision to write helix-core in **Rust not Go** (D2 [SYNTHESIS §3]) rests on staying under it with ≥30% headroom. MEM also covers: no RSS growth over a 24 h soak (no leak in the reconcile loop or the QUIC buffer pool), the coordinator's bounded memory at 10k streams (SLO4 §7.2), and Android VpnService RSS.

Harness. iOS: Instruments Allocations/Leaks + footprint against a device build, peak-RSS series captured as a .trace; a soak harness runs a sustained transfer for 24 h sampling /proc/<pid>/status (Linux core), dumpsys meminfo

(Android), Instruments (iOS). The G3 gate: peak RSS < ceiling, headroom $\geq 30\%$, proven on a **real device** (a simulator does not reproduce the jetsam ceiling — §11.4.3: no device $\Rightarrow$ honest SKIP: hardware_not_present, never a fake PASS).

```
// shims/apple/PacketTunnelProvider+MemoryProbe.swift (G3
// evidence emitter)
func sampleFootprint() -> UInt64 { // bytes; written
    to qa-results/mem/ios_rss.csv
    var info = task_vm_info_data_t()
    var count =
        mach_msg_type_number_t(MemoryLayout<task_vm_info>.size /
        MemoryLayout<integer_t>.size)
    let kr = withUnsafeMutablePointer(to: &info) {
        $0.withMemoryRebound(to: integer_t.self, capacity:
        Int(count)) {
            task_info(mach_task_self_,
            task_flavor_t(TASK_VM_INFO), $0, &count) } }
    return kr == KERN_SUCCESS ? info.phys_footprint : 0 //
    phys_footprint = the jetsam metric
}
```

Evidence. RSS-vs-time CSV + Instruments .trace; G3 verdict = peak < ceiling $\times$ 0.7. **Gate.** Phase-0 G3 (release-blocking make-or-break); MVP soak for SLO4.

5.13 BENCH + PERF + SCALE

Covers. BENCH: raw throughput ($\geq 80\%$ bare-link plain-UDP G1; $\geq 50\%$ of plain over MASQUE G2 [04_P0]), CPU-per-Gbps for both candidate edge languages (the Go-vs-Rust G4 decision input, D5 [SYNTHESIS]), handshakes/sec, MASQUE framing latency. **PERF:** the four SLOs as p99 budgets (§7.2) — event $\rightarrow$ delta < 1 s, enroll $\rightarrow$ first-map < 2 s, revoke $\rightarrow$ enforcement < 1 s; Go GC-tail watch on the coordinator hot path. **SCALE:** N simulated agents holding WatchNetworkMap streams; convergence p99 < 1 s and bounded coordinator memory over a 24 h soak (SLO4).

Harness. bench.sh [04_P0 §8] driving the netns rig with iperf3 -J + per-iteration CSV, run N=3 deterministic (§11.4.50). criterion for Rust micro-benches (framing). A coordinator agent-fuzzer holds N streams and measures convergence + RSS.

```
# bench.sh - the G1/G2/G4 benchmark, deterministic over N=3 [04_P0
# §8, 06 HVPN-P0-NNN]
THRPUT=$(iperf3 -J -c 10.10.0.20 -P "$CLIENTS" | jq
'.end.sum_received.bits_per_second')
GOODPUT_LOSS=$(with_netem 'loss 5% delay 40ms'
iperf3_goodput) # G2 / resilience
CPU_PER_GBPS=$(awk -v t="$THRPUT" -v c="$EDGE_CPU_SEC"
'BEGIN{print c/(t/1e9)}') # G4 input
printf '%s,%s,%s,%s,%s\n' "$EDGE_LANG" "$RUN" "$THRPUT"
"$GOODPUT_LOSS" "$CPU_PER_GBPS" \
>> qa-results/bench/edge_compare.csv # the G4 decision CSV
(rust vs go)
```

Evidence. the per-metric CSV with min/max/mean/p95 (§11.4.24 resource-stats discipline); the G4 CSV decides gates.G4.outcome IN ('rust', 'go'). **Gate.** Phase-0 G1/G2/G4; MVP SLO PERF/SCALE gates; a PERF regression > budget is a release blocker.

5.14 UI + UX + REC — recorded-evidence (vision_engine / panoptic)

Covers. The three Flutter flavors (Access / Connector / Console [04_UI]): **UI** = widget + golden tests on the design-system components (ConnectButton, StatusChip, ExitPicker, ShieldIndicator) across light+dark (§11.4.162 OpenDesign parity); **UX** = the real user journey (enroll → connect → reach → see the shield go green) driven through the app's own UI; **REC** = a **window-scoped MP4** (§11.4.154/.155 — app window only, never whole desktop) of that journey, fed to the media-validation pipeline (§11.4.163) for an OCR/vision verdict.

Harness. flutter test (widget + golden). panoptic (UI capture harness, Go [05 §6]) drives the web/desktop/mobile app and captures the window-scoped MP4; vision_engine (Go) runs OCR/state-verify on the frames — e.g. it reads that the ConnectButton text transitions Connect → Connecting → Connected and the ShieldIndicator turns green, then cross-checks against the *real* core status stream (a green shield while the core FSM is Blocked is B3 and FAILs). Filenames carry the §11.4.155 project prefix helixvpn---connect-flow---<run-id>.mp4; raw corpus is git-ignored, curated evidence committed under docs/qa/<run-id>/ (§11.4.83).

```
// helix-ui/test/connect_button_golden_test.dart (UI -
    light+dark, §11.4.162)
void main() {
  for (final brightness in [Brightness.light, Brightness.dark]) {
    testWidgets('ConnectButton golden - $brightness', (tester) {
      async {
        await tester.pumpWidget(HelixTheme(brightness: brightness,
          child: ConnectButton(state: ConnState.connected)));
        await expectLater(find.byType(ConnectButton),
          matchesGoldenFile('goldens/
            connect_button_connected_$brightness.png'));
      });
    }
  }

  // the §11.4.163 media-validation verdict vision_engine emits for
  // the REC artifact
  { "artifact": "docs/qa/<run-id>/helixvpn---connect-flow---<run-
    id>.mp4",
    "expected_patterns":
    ["Connect","Connecting","Connected","shield:green"], // SPECIFY-
    phase (§11.4.159(J))
    "matched": ["Connect","Connecting","Connected","shield:green"],
    "cross_check": { "core_fsm_at_green_shield":
    "Connected" }, // defeats B3
    "verdict": "PASS", "self_validated":
    true } // golden-bad MP4 → FAIL
  (§3.3)
```

Evidence. golden PNGs + the window-scoped MP4 + the vision verdict JSON.
Gate. make test UI stage + make qa REC stage; AC9 (three apps drive it). A user-visible item without a vision-verified REC is a §11.4.153 release blocker.

6. The coverage ledger (feature × test-type × evidence-state)

The coverage ledger is the §11.4.25/.52/.153 mechanical answer to "is every feature proven, by every warranted test type, with real evidence?" It is a **git-tracked SQLite table** (§11.4.93/.95) — a projection of the workable-items DB cross-joined with the test-type matrix — regenerated by doc_processor from the spec feature map (§11.4.153) on every QA run, and rendered to docs/qa/coverage_ledger.md (+HTML/PDF) by docs_chain (§11.4.106).

6.1 DDL

```
-- docs/.workable_items.db (git-tracked, §11.4.95) – coverage
ledger projection
CREATE TABLE IF NOT EXISTS features (
  feature_id TEXT PRIMARY KEY, -- 'F-
    KILLSWITCH', 'F-AUTHZ-REACH'
  component TEXT NOT NULL, -- 'helix-
    core' | 'helix-go' | 'helix-edge' | 'helix-ui'
  title TEXT NOT NULL CHECK (length(title) >= 40), --
    §11.4.91
  evidence_class TEXT NOT NULL, -- §11.4.69
    class (network_connectivity, ...)
  phase TEXT NOT NULL CHECK (phase IN
    ('P0', 'P1', 'P2', 'P3'))
);
CREATE TABLE IF NOT EXISTS coverage_cells (
  feature_id TEXT NOT NULL REFERENCES features(feature_id),
  test_type TEXT NOT NULL, -- §2 abbrev:
    UNIT|INT|E2E|SEC|...
  state TEXT NOT NULL CHECK (state IN
    ('REQUIRED', 'AUTONOMOUS_VERIFIED', 'AUTONOMOUS_DESIGNED',
    'OPERATOR_ATTENDED_ONLY', 'SKIP', 'NOT_APPLICABLE', 'FAIL', 'PENDING')),
  skip_reason TEXT, -- closed set
    (§11.4.3) when state IN (SKIP, NOT_APPLICABLE)
  evidence_path TEXT, -- §11.4.69 –
    REQUIRED for *_VERIFIED states
  last_run_at TEXT,
  diary_ref TEXT, -- §11.4.149
    test_diary row id
  PRIMARY KEY (feature_id, test_type),
  -- §11.4.69: a VERIFIED cell MUST carry an evidence path
    (schema makes a bluff impossible)
  CHECK (state NOT IN ('AUTONOMOUS_VERIFIED') OR (evidence_path
    IS NOT NULL AND length(evidence_path)>0)),
  -- §11.4.3: a SKIP/NA cell MUST carry a closed-set reason
```

```

        CHECK (state NOT IN ('SKIP', 'NOT_APPLICABLE') OR (skip_reason
            IS NOT NULL))
    );
CREATE VIEW v_coverage_gaps AS
    SELECT f.feature_id, c.test_type, c.state
    FROM features f JOIN coverage_cells c USING (feature_id)
    WHERE c.state IN ('PENDING', 'FAIL', 'OPERATOR_ATTENDED_ONLY');
    -- the release-blocker set

```

6.2 State machine for a cell

stateDiagram-v2

```

    [*] --> REQUIRED: feature declares the type (§11.4.169)
    REQUIRED --> AUTONOMOUS_DESIGNED: test authored, RED on broken
    artifact (§11.4.115)
    AUTONOMOUS_DESIGNED --> AUTONOMOUS_VERIFIED: GREEN +
    evidence_path captured (§11.4.69)
    AUTONOMOUS_VERIFIED --> FAIL: re-run regresses (§11.4.4 STOP)
    FAIL --> AUTONOMOUS_DESIGNED: systematic-debug + fix (§11.4.102)
    REQUIRED --> SKIP: topology/hardware absent (§11.4.3 closed
    reason)
    REQUIRED --> NOT_APPLICABLE: e.g. DDOS@MVP single-node (§11.4.6)
    AUTONOMOUS_VERIFIED --> [*]: feature closeable when ALL required
    cells VERIFIED/NA

```

Release rule. A phase ships only when v_coverage_gaps is empty for that phase (every required cell is AUTONOMOUS_VERIFIED or NOT_APPLICABLE with a reason); OPERATOR_ATTENDED_ONLY is a release blocker until promoted with a tracked migration item (§11.4.52). The ledger is the §11.4.40 full-suite-retest scoreboard.

6.3 Worked rows (illustrative)

feature_id	UNIT	INT	E2E	SEC	STRESS	CHAOS	CONC	RACE	MEM	BE
F-AUTHZ-REACH	✓	✓	✓	✓	✓	✓	—	—	—	✓
F-DEFAULT-DENY	✓	✓	✓	✓	—	✓	—	✓	—	—
F-KILLSWITCH	✓	—	✓	✓	—	✓	—	✓	—	—
F-TRANSPORT-ESCALATE	✓	✓	✓	✓	✓	✓	—	✓	—	✓
F-POLICY-RECONCILE	✓	✓	✓	—	✓	✓	✓	✓	—	✓
F-REVOKE	✓	✓	✓	✓	—	✓	✓	—	—	✓
F-IOS-NE-MEM	—	—	—	—	—	—	—	—	✓(G3)	—
F-NO-LOG-SCHEMA	✓	✓	—	✓	—	—	—	—	—	—

feature_id	UNIT	INT	E2E	SEC	STRESS	CHAOS	CONC	RACE	MEM	BE
F-DDOS-FLOOD	—	—	—	NA ¹	—	—	—	—	—	—

¹ NOT_APPLICABLE: single-node-selfhost at MVP; re-arms Phase 2 (§5.8).

7. Per-phase acceptance gates

7.1 Phase 0 — Spike exit gates (G1–G6) [04_P0, 06]

Gate	Question	Go/No-Go bar	Evidence	Test types
G1	plain-UDP WG client→gw→connector LAN?	iperf3 ≥ 80% bare-link; curl 10.10.0.20:80 OK	pcap + iperf3 CSV	E2E, BENCH, SC, FA, CH
G2	same core over MASQUE/QUIC through a DPI UDP block?	≥ 50% of plain-UDP; survives 5% loss > UoT strawman; tshark classifies HTTP/3, no WG sig	pcap + tshark + CSV	E2E, SEC, BENCH, FA
G3	iOS NE Rust core under memory ceiling? (make-or-break)	peak RSS < ceiling, ≥30% headroom on a real device	Instruments .trace + RSS CSV	MEM
G4	Go-vs-Rust edge benchmark?	decision recorded from CPU-per-Gbps/p99/churn/mem CSV (N=3)	edge_compare.csv	BENCH, PERF, RACE
G5	flutter_rust_bridge FFI drives core from Dart?	Dart calls connect(), receives status stream	FFI round-trip log + UI recording	UNIT, UI, FA
G6	push-based reconcile from a static map?	a map edit reconciles the slice with no restart	reconcile event log	INT, E2E, FA

Each gate is a row in the gates table ([06 §0.3]); outcome flips to pass/fail/rust/go only with an evidence_path. A gate that cannot clear inside the time box **is the finding** — escalate the decision (§11.4.66), never overrun silently.

7.2 Phase 1 — MVP Definition-of-Done (AC1–AC9 + SLO1–SLO4) [04_P1, 07]

The MVP ships only when **all nine ACs and all four SLOs** pass on a clean baseline (§11.4.40 retest + §11.4.108 runtime-signature on a fresh deploy). Each AC is a challenges bank entry (§5.5) scored by `helix_qa` (§5.6).

AC	Criterion	Captured evidence	Bound item
AC1	self-host from zero (<code>helixvpnctl init + start</code>)	recorded terminal, healthchecks green, <code>helix_up==1</code>	HVPN-P1-...
AC2	enroll connector+client; reach authorized LAN host	E2E netns capture + pcap (body hash)	F-AUTHZ-REACH
AC3	DENY unauthorized host (default-deny)	negative E2E: SYN out, zero SYN-ACK	F-DEFAULT-DENY
AC4	auto-escalate to MASQUE when WG blocked, stays up	<code>StatusReport.transport=="masque-h3"</code> + <code>tshark H3</code>	F-TRANSPORT-ESCALATE
AC5	policy edit reconfigures all devices < 1 s, no restart	<code>helix_reconcile_seconds p99 < 1 s</code>	F-POLICY-RECONCILE
AC6	revoke device < 1 s	<code>revoke→edge-enforcement</code> timing CSV	F-REVOKE
AC7	kill-switch + DNS-leak: no plaintext on drop	host pcap, zero leak / zero :53 (§5.7)	F-KILLSWITCH
AC8	no durable conn/traffic log (schema-lint green)	<code>schemalint PASS</code> + §1.1 mutation FAIL	F-NO-LOG-SCHEMA
AC9	three apps drive it all (Access+Connector+Console)	per-app window-scoped UX recordings + vision verdict	F-UI-FLOWS

SLO	Target	Test type	Item
SLO1 event→delta-on-wire	<code>p99 < 1 s</code>	PERF	HVPN-P1-073
SLO2 enroll→first NetworkMap	< 2 s	PERF	HVPN-P1-031
SLO3 revoke→edge enforcement	< 1 s	PERF	HVPN-P1-033
SLO4 coordinator memory @ 10k streams	bounded / 24 h soak	MEM, SCALE	HVPN-P1-074

7.3 Phase 2 / 3 — re-armed and new gates

Phase 2 **re-arms** DDOS + full SCALE (HA multi-region, stateless coordinators, Patroni PG, NATS JetStream) and adds: DAITA traffic-shaping evidence (maybenot), P2P NAT-traversal E2E (hole-punch success rate over a STUN/DERP rig), multi-hop nested-WG E2E, post-quantum handshake interop (ML-KEM hybrid, never PQ-only). Phase 3 adds HarmonyOS NEXT + Aurora OS platform shim E2E (the biggest platform risk — real native tunnel-shim work), WASM browser-MASQUE proxy E2E, and a third-party audit + reproducible-build attestation. Each re-arm flips its coverage-ledger cells from NOT_APPLICABLE/PENDING to REQUIRED mechanically (§6.2).

8. Execution discipline: determinism, risk-order, four-layer, diary

Determinism (§11.4.50). Every PASS runs $N=3$ (normal) / $N=10$ (cycle-validation) against the same artifact MD5 + same rig, identical evidence-hashes; a divergent run is auto-FAIL, no flake escape (`ab_run_n_times`, §5.4).

Risk-ordering (§11.4.132). The suite runs **highest-risk first**: (a) most-recently-worked, (b) historically most-problematic, (c) highest crash/leak likelihood, (d) most-reopened (`reopens_count` from the workable-items DB). For HelixVPN the irreversible-security floor — kill-switch (AC7), default-deny (AC3), RLS (S9), revoke (AC6), key-never-leaves (S2) — runs **before** any convenience/UI test, and only after that set is GREEN with captured evidence does the rest run.

Four-layer per closure (§11.4.4(b)). Every workable item closes only with: (1) a pre-build gate (lint/schema/sh -n/self-validated-analyzer present), (2) a post-build/runtime test on the deployed artifact, (3) a runtime-signature on a **clean** deploy (§11.4.108/.139 — never validate against a stale daemon, defeats B4), (4) a paired §1.1 mutation that makes the gate FAIL (proves the gate is not a bluff).

Reproduce-first + extend (§11.4.146). Every fix authors a RED test that reproduces the defect on the *current broken* artifact (`RED_MODE=1`), flips to a GREEN guard (`RED_MODE=0`) on the fix, then **fans out** across the feature's full case-space (valid/invalid/boundary/concurrent/chaos) and registers a permanent regression guard (§11.4.135).

Test diary (§11.4.149). Every run appends a row to the item's `test_diary` (`date · tested_by ∈ {AI-agent, HelixQA, Operator} · result · observations · evidence_path · feature_class`). The schema constraint **rejects a PASS row with an empty evidence_path** — a diary-layer bluff is impossible.

```
CREATE TABLE IF NOT EXISTS test_diary (           -- §11.4.149, git-
    tracked
    diary_id      INTEGER PRIMARY KEY,
    atm_id        TEXT NOT NULL REFERENCES items(atm_id),
    date_time     TEXT NOT NULL,                  -- ISO-8601 UTC
    tested_by     TEXT NOT NULL CHECK (tested_by IN ('AI-
        agent', 'HelixQA', 'Operator', 'User')),
    result        TEXT NOT NULL CHECK (result IN
        ('PASS', 'FAIL', 'SKIP')),
    observations  TEXT NOT NULL,
    evidence_path TEXT, feature_class TEXT,
    CHECK (result <> 'PASS' OR (evidence_path IS NOT NULL AND
        length(evidence_path) > 0)) -- §11.4.69
);
```

Operator-escape ⇒ bluff-audit (§11.4.138). If the operator finds a defect the green suite missed, that is by definition a PASS-bluff: it triggers a systematic-debug pass, a bluff-audit naming the exact assertion that should have caught it (`file:line`), a permanent regression guard in the same commit, and the audit committed under `docs/research/<scope>/<defect>_bluff_audit/`.

9. The local gate harness (Makefile + git hooks, §11.4.75/.156)

All enforcement is **local** (§11.4.156 — no active remote CI). The seam:

```
# Makefile – the local gate harness (excerpt; full set in [05])
gen:          ; buf generate && flutter_rust_bridge_codegen
              generate      # proto + FFI
build:        ; cargo build && go build ./... && melos run build
test:         ; cargo test && go test -race ./helix-go/... && melos
              run test     # boots infra via containers (§5.2)
e2e:          ; sudo rig/netns_up.sh && rig/test_reach.sh pass &&
              rig/test_reach.sh deny ; sudo rig/netns_down.sh
sec:          ; rig/killswitch_drop.sh && bash scripts/
              secret_audit.sh      # §5.7
chaos:        ; bash scripts/
              stress_chaos.sh      # §5.9
bench:        ; bash bench.sh
              3                    # N=3
              deterministic (§5.13)
qa:           ; go run ./submodules/helix_qa/cmd/orchestrator --
              suite mvp-dod      # 8 ACs, captured evidence (§5.6)
coverage:     ; go run ./tools/coverage_ledger && docs_chain sync --
              context .docs_chain/contexts/helixvpn_spec.yaml
docs-verify;; docs_chain verify --context .docs_chain/contexts/
              helixvpn_spec.yaml # deterministic gate (§11.4.106)
```

Git-hook layers (§11.4.75): pre-commit runs UNIT + sh -n + schema-lint + no-log lint + mutation-residue scan (§11.4.84); pre-push runs INT (containers boots PG+Redis) + the SEC subset + propagation gates; post-commit auto-exports HTML/PDF/DOCX siblings (§11.4.65/.153). The qa.yml.disabled-local-only file is preserved but inert (§11.4.75 Layer-5). Long suites run backgrounded (nohup ... & disown, §11.4.89) so the main work stream is never blocked.

Container-test infra is rootless (§11.4.161). containers/pkg/boot runs Podman rootless; no sudo, no rootful Docker — the only sudo in the whole harness is rig/netns_up.sh (network-namespaces creation needs CAP_NET_ADMIN), and that is a documented, scoped exception, never a container-management escalation.

10. Open decisions surfaced for QA

Per §11.4.66, decisions the QA layer touches but does not silently resolve:

#	Decision	Options	Recommendation
QA-D1	Edge language (G4) drives which RACE toolchain is canonical	Rust (loom+tsan) vs Go (-race)	Rust — loom's exhaustive interleaving is stronger for the verdict-swap hot path [04_P0 G4]; revisit on the G4 CSV.

#	Decision	Options	Recommendation
QA-D2	DDoS suite timing	author-now-parked vs defer-authoring	Author-now-parked — build the bank in MVP so it re-arms mechanically in Phase 2; the seam (rate-limit, fail-static) exists now [04 §T9].
QA-D3	iOS MEM gate on CI-less hosts	real-device-required vs simulator-allowed	Real-device-required — the jetsam ceiling does not reproduce in the simulator; no device ⇒ honest SKIP: <code>hardware_not_present</code> , never a sim PASS (§11.4.3).
QA-D4	Vision verdict authority	trust harness verdict vs independent <code>vision_engine</code> re-read	Independent re-read — <code>vision_engine</code> re-reads the MP4 and cross-checks the core FSM (defeats B3); the harness's claim is never the verdict (§11.4.165).
QA-D5	Challenge engine vs raw test exit codes for DoD	exit-code vs challenges-scored	challenges-scored — the engine re-reads the evidence artifact, structurally defeating B1–B3; exit codes alone are bluffable (§11.4.27).

Sources

- [04_P0] 04_VPN_CLD/HelixVPN-Phase0-Spike.md — netns+nftables+netem rig, G1–G6 gates, bench.sh, surviving interfaces, iOS memory gate.
- [04_P1] 04_VPN_CLD/HelixVPN-Phase1-MVP.md — MVP DoD (AC1–AC9), SLOs, no-log schema-lint, RLS, coordinator/WatchNetworkMap.
- [04_ARCH] 04_VPN_CLD/HelixVPN-Architecture-Refined.md — three roles, transport stack, security invariants.
- [04_UI] 04_VPN_CLD/HelixVPN-helix-ui-Flutter.md — Flutter flavors, design system, UI/golden test surface.
- [05_YBO] founding constraints + mandated stack. [02_QWN] [01_DSK] [11_MST] [10_KMI] [07_GMI] — LLM analyses (transport plurality, IPAM/D4 solvers, asymmetric per-leg transport).
- [SYNTHESIS] v09-research/_SYNTHESIS.md — settled floor, key decisions D1–D7, phased structure, ecosystem-integration gaps.
- [01] [02] [03] [04] [05] [06] [07] — companion final/ documents (data plane, control plane, client core+UI, security/PKI, repo+ecosystem, Phase-0 WBS, Phase-1 WBS).
- External facts: [research-masque] RFC 9297 (HTTP Datagrams)/9298 (CONNECT-UDP)/9221 (unreliable datagram); [research-boringtun], [research-quinn] (Rust QUIC), [research-wireguard] (Noise IK).
- Constitution: §11.4.169 (test-type mandate), §11.4.27 (no-fakes/100% type coverage), §11.4.5/.69/.107 (captured evidence / sink-side / liveness), §11.4.85 (stress+chaos), §11.4.50 (determinism), §11.4.132 (risk-order), §11.4.4(b)/.108/.139/.146/.135 (four-layer / runtime-signature / reproduce-first / regression guard), §11.4.149 (test diary), §11.4.25/.52/.153 (coverage ledger),

§11.4.76/.161 (containers/rootless), §11.4.156 (no active remote CI), §11.4.75 (local hook layers), §11.4.159/.163/.165 (recorded evidence / media validation / independent verification), §11.4.93/.95 (workable-items DB).